

Inverse functions

An inverse reverses a function's input-output pairs.

If $f(2) = 7$, what does its inverse do to 7?

One-to-one functions

A function is one-to-one when different inputs always give different outputs.

$$f(x) = x^2, \quad x \in \mathbb{R}$$

Why does this function need a domain restriction to have an inverse function?

The horizontal line test

A graph represents a one-to-one function when every horizontal line meets it at most once.

Does a cubic function $f(x) = x^3$ pass this test?

An odd-power inverse

$$f(x) = x^3 + 2$$

What is the inverse function?

A scaled cubic

$$f(x) = 2(x - 1)^3 + 5$$

What is the inverse function and its domain?

The inverse notation

$$f^{-1}(x) \quad \text{and} \quad \frac{1}{f(x)}$$

Why do these expressions generally describe different functions?

Restricting a square

$$f(x) = x^2, \quad x \geq 0$$

What inverse function results from this restriction?

The other square branch

$$f(x) = x^2, \quad x \leq 0$$

What inverse function results from this restriction?

A shifted quadratic

$$f(x) = (x - 3)^2 + 1, \quad x \geq 3$$

What is the inverse function?

Inverse domain and range

$$f(x) = (x - 3)^2 + 1, \quad x \geq 3$$

What are the domain and range of its inverse?

Choosing a restriction

$$f(x) = 2(x + 1)^2 - 4$$

Which two maximal intervals each make f one-to-one?

A decreasing quadratic branch

$$f(x) = 2(x + 1)^2 - 4, \quad x \leq -1$$

What is the inverse on this restricted domain?

Verifying an inverse

Use the stated domains.

$$f(x) = x^2, \quad x \geq 0; \quad g(x) = \sqrt{x}$$

How can both compositions verify that these functions are inverses?

Why the restriction matters

$$\sqrt{x^2} = |x|$$

Why does this identity prevent the square root from undoing squaring on all real inputs?

Graph reflection

Graphs of inverse functions reflect across the line $y = x$.

Where does the point (4, 9) appear on the inverse graph?

A radical function's domain

$$g(x) = \sqrt{2x - 6} + 1$$

What are the real domain and range?

Inverting a square-root function

$$f(x) = \sqrt{x - 2} + 3$$

What is its inverse, including the inverse domain?

An invertible rational function

$$f(x) = \frac{x + 1}{x - 2}$$

What inverse formula and excluded input result?

Checking the rational inverse domain

$$f(x) = \frac{x + 1}{x - 2}$$

Why can the inverse not accept the input 1?

A geometric inverse

The volume in cubic centimeters of a cube with side length s cm is $V(s) = s^3$, with $s > 0$.

What inverse function gives side length from volume?

Restricting to physical inputs

The area in square meters of a square is $A(s) = s^2$, with $s > 0$.

Which inverse branch gives a physical side length?

Preparing to invert an exponential

$$f(x) = 2^x$$

Does this function repeat an output, and why does that matter for an inverse?